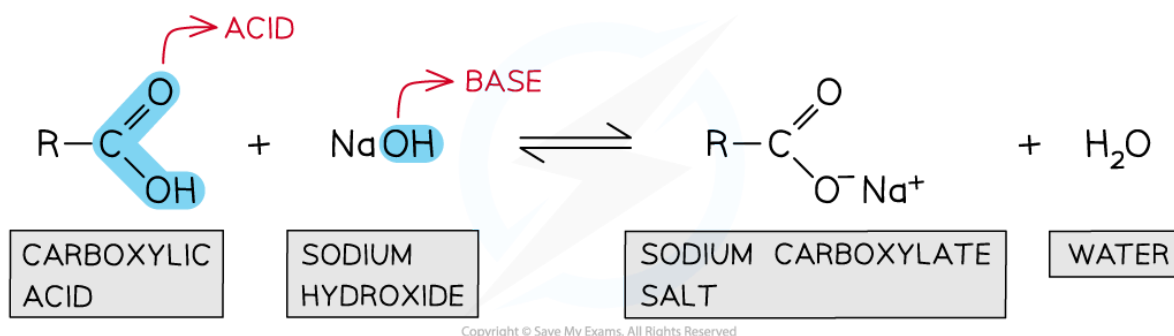


Relative Acidities of Carboxylic Acids, Phenols & Alcohols (Cambridge (CIE) A Level Chemistry): Revision Note

Relative Acidities of Carboxylic Acids, Phenols & Alcohols

- **Carboxylic acids** are compounds with a -COOH functional group
- They can act as **acids** and lose a proton (H^+ ion) in an aqueous solution to form **carboxylate salts** and **water**

Carboxylic acids forming carboxylate salts



Carboxylic acids dissociate in aqueous solutions to form carboxylate salts and water

- However, carboxylic acids are only **weak acids** as the **position of equilibrium** lies well over to the **left-hand side**
- The $\text{p}K_a$ values of carboxylic acids, phenols, and alcohols suggest that carboxylic acids are **stronger acids** than alcohols and phenols
 - The $\text{p}K_a$ is a measure of the relative strength of a species as an acid
 - The smaller the $\text{p}K_a$ value, the stronger the acid

Relative acidity of ethanol, phenol & carboxylic acids table

Acid	Dissociation	$\text{p}K_a$ at 25 °C
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Ethanol	$\text{C}_2\text{H}_5\text{OH (aq)} \rightleftharpoons \text{C}_2\text{H}_5\text{O}^- \text{ (aq)} + \text{H}^+ \text{ (aq)}$	16
Phenol	$\text{C}_6\text{H}_5\text{OH (aq)} \rightleftharpoons \text{C}_6\text{H}_5\text{O}^- \text{ (aq)} + \text{H}^+ \text{ (aq)}$	10
Ethanoic acid	$\text{CH}_3\text{COOH (aq)} \rightleftharpoons \text{CH}_3\text{COO}^- \text{ (aq)} + \text{H}^+ \text{ (aq)}$	4.8
Benzoic acid	$\text{C}_6\text{H}_5\text{COOH (aq)} \rightleftharpoons \text{C}_6\text{H}_5\text{COO}^- \text{ (aq)} + \text{H}^+ \text{ (aq)}$	4.2

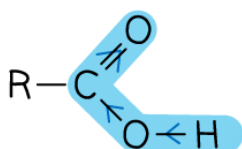
- This order of relative acidities can be explained by looking at the **strength** of the O-H bond and the stability of the **conjugate bases** of the acids

Strength of O-H bond

- In carboxylic acids, the electrons in the O-H bond are drawn towards the C-O bond
- The electrons in the C-O bond are drawn towards the C=O bond
- Overall, the O-H bond is **weakened** due to the **carbonyl** (C=O) group removing electron density from it and drawing it towards itself
- Carboxylic acids can therefore more easily **lose** a proton compared to phenols and alcohols which **lack** this electron-withdrawing carbonyl group

Comparing OH bond strength of carboxylic acids, ethanol and phenol

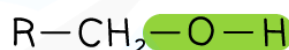
CARBOXYLIC ACIDS



ELECTRONS ARE DRAWN TOWARDS C=O CAUSING THE O-H BOND TO BECOME WEAKER;

IT'S EASIER TO ACT AS AN ACID AND LOSE H^+

ETHANOL



NO C=O GROUP SO O-H BOND IS NOT WEAKENED;
IT'S HARDER TO ACT AS AN ACID AND LOSE H^+

PHENOL

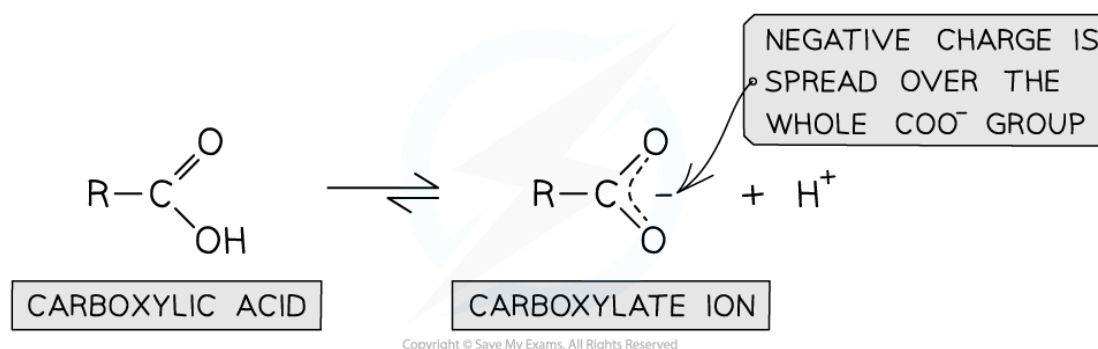


The carbonyl group in carboxylic acids draws the electrons away from the O-H bond causing it to become weaker compared to the O-H bond in phenols and alcohols

Stability of carboxylate ions

- The **conjugate base** of carboxylic acids is the **carboxylate ion**
- The **charge density** on the oxygen atom is **spread out** over the carboxylate ion
- This is because the charge is **delocalised** on an **electronegative carbonyl oxygen atom**
- As a result, the electrons on the oxygen atom are **less available** for bond formation with an H^+ ion to reform the **undissociated** acid molecule with $-\text{COOH}$ group
- The position of the dissociation equilibrium lies more to the right compared to alcohols and phenols

The equilibrium position of a carboxylic acid and its carboxylate ion



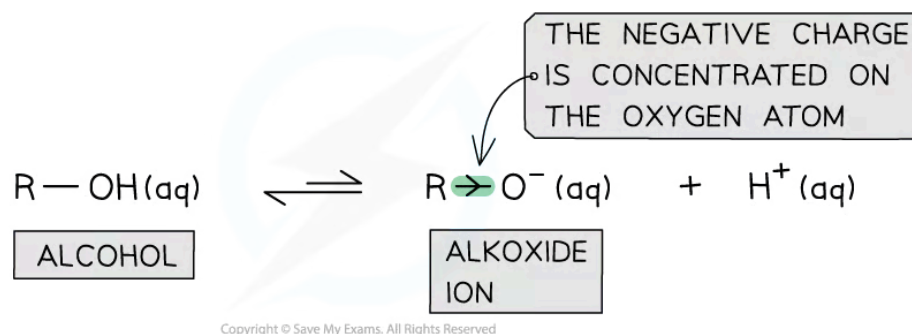
The carboxylate ion is stable due to the delocalisation of the charge density on the electronegative oxygen

Stability of alkoxide ions

- The **conjugate base** of alcohols is the **alkoxide ion**
- The **alkyl** group in the ion is an **electron-donating** group that donates electron density to the oxygen atom
- As a result, the electron density on the oxygen atom is **more readily available** for bond formation with an H^+ ion
- Alkoxide ions also **lack** the ability to delocalise the charge density on the entire ion
- The conjugate bases of alcohols are therefore less stable than the alcohols themselves and are more likely to reform the alcohol

- This means that alcohols are **weaker acids** compared to carboxylic acids and phenols
- The position of the dissociation equilibrium lies more to the left

The equilibrium position of an alcohol and its alkoxide ion

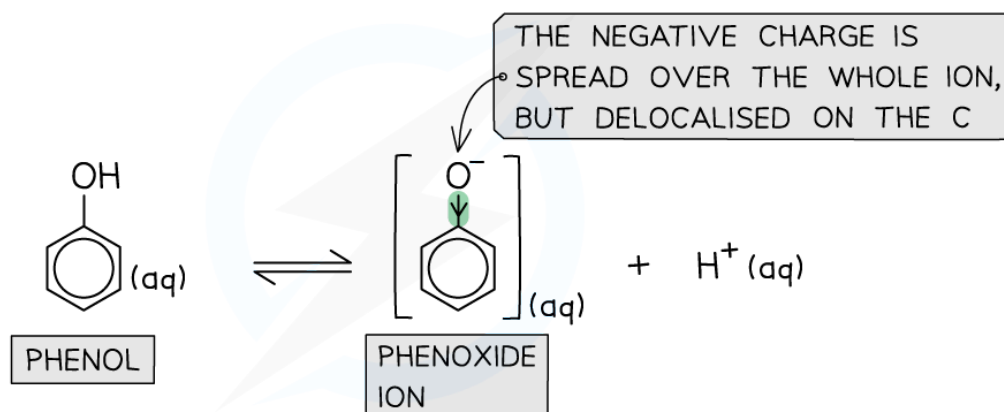


The electron-donating alkyl groups in alkoxide ions increase the electron density on the oxygen atom which is, therefore, more likely to bond with a H^+ ion and reform the alcohol

Stability of phenoxide ions

- In the **phenoxide ion** (which is the conjugate base of phenol) the charge density on the oxygen atom is **spread out** over the entire ion
 - This delocalisation of electrons **stabilises** the phenoxide ion
- As a result, the electrons on the oxygen atom are **less available** for bond formation with a proton (H^+ ion)
- The **conjugate base** of phenols is therefore **more stable** than phenol
- However, since the delocalisation of charge density is on carbon atoms and not on electronegative oxygen atoms like in the carboxylate ion, phenoxide ions are **less stable** than carboxylate ions
- Therefore, phenols are **weaker acids** relative to carboxylic acids
- The position of the dissociation equilibrium lies more to the right compared to alcohols and more to the left compared to carboxylic acids

The equilibrium position of phenol and the phenoxide ion



The charge density is delocalised on the entire benzene ring in the phenoxide ions